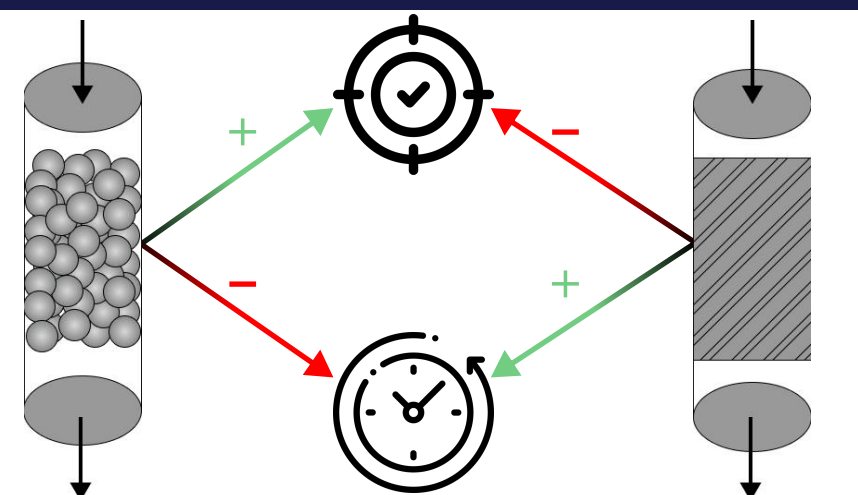


Motivation

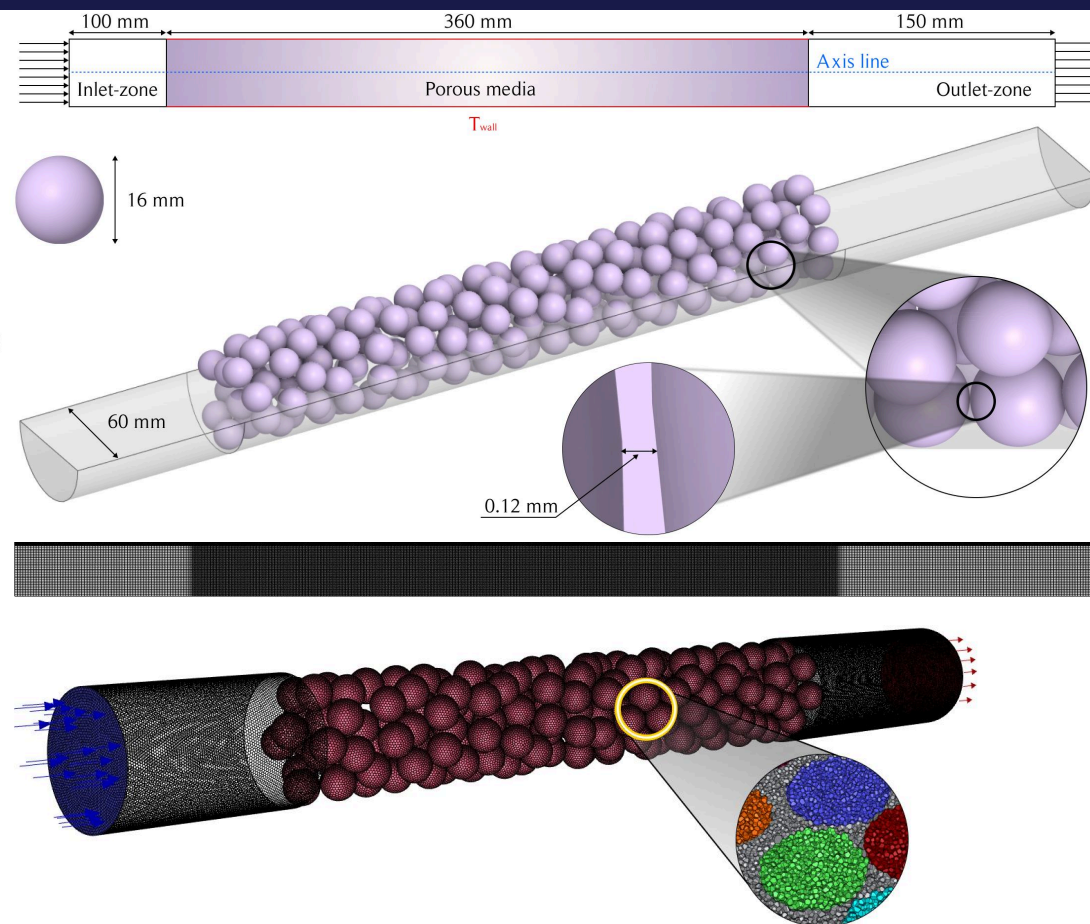


The Particle-Resolved Model (PRM) considers each individual catalyst particle within a packed bed reactor.

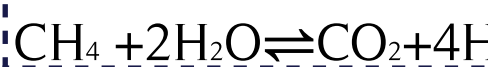
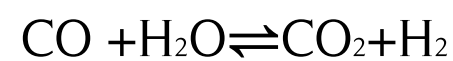
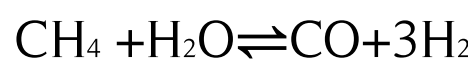
The Pseudo-homogeneous Model (PHM) considers packed bed reactor as a porous medium.

The objective is to conduct a comparative analysis of the PRM and PHM models for steam methane reforming (SMR) performance.

Methods



Reaction kinetics:



Diffusion model:

$$D_i^{\text{eff}} = \left(\frac{1}{D_{m,i}^{\text{eff}}} + \frac{1}{D_{i,k}^{\text{eff}}} \right)^{-1}$$

$$D_{m,i}^{\text{eff}} = \frac{\epsilon}{\tau} \sum_{i=1}^{\infty} \frac{1 - x_i}{x_i / D_{ij}}$$

$$D_{ij} = \frac{0.0143T^{1.75}}{p_i M^{0.5} (V_i^{1/3} + V_j^{1/3})^2}$$

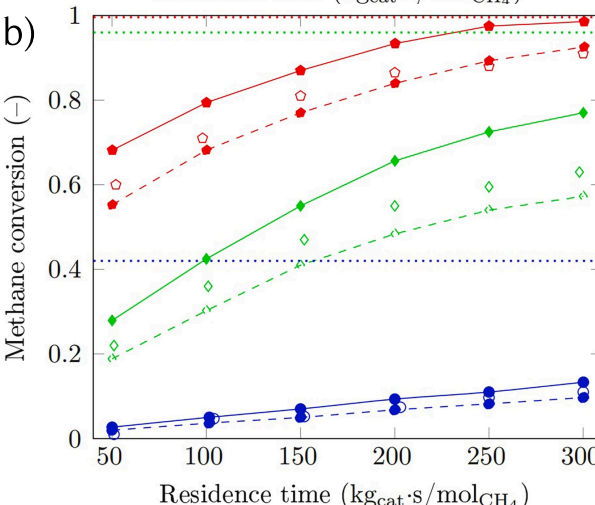
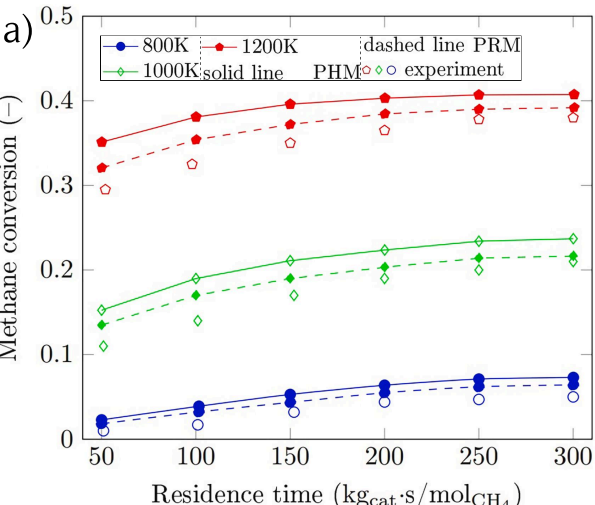
$$D_{i,k}^{\text{eff}} = \frac{\epsilon}{\tau} \cdot 48.5 \cdot d_p \left(\frac{T}{M_i} \right)^{0.5}$$

Effectiveness factor:

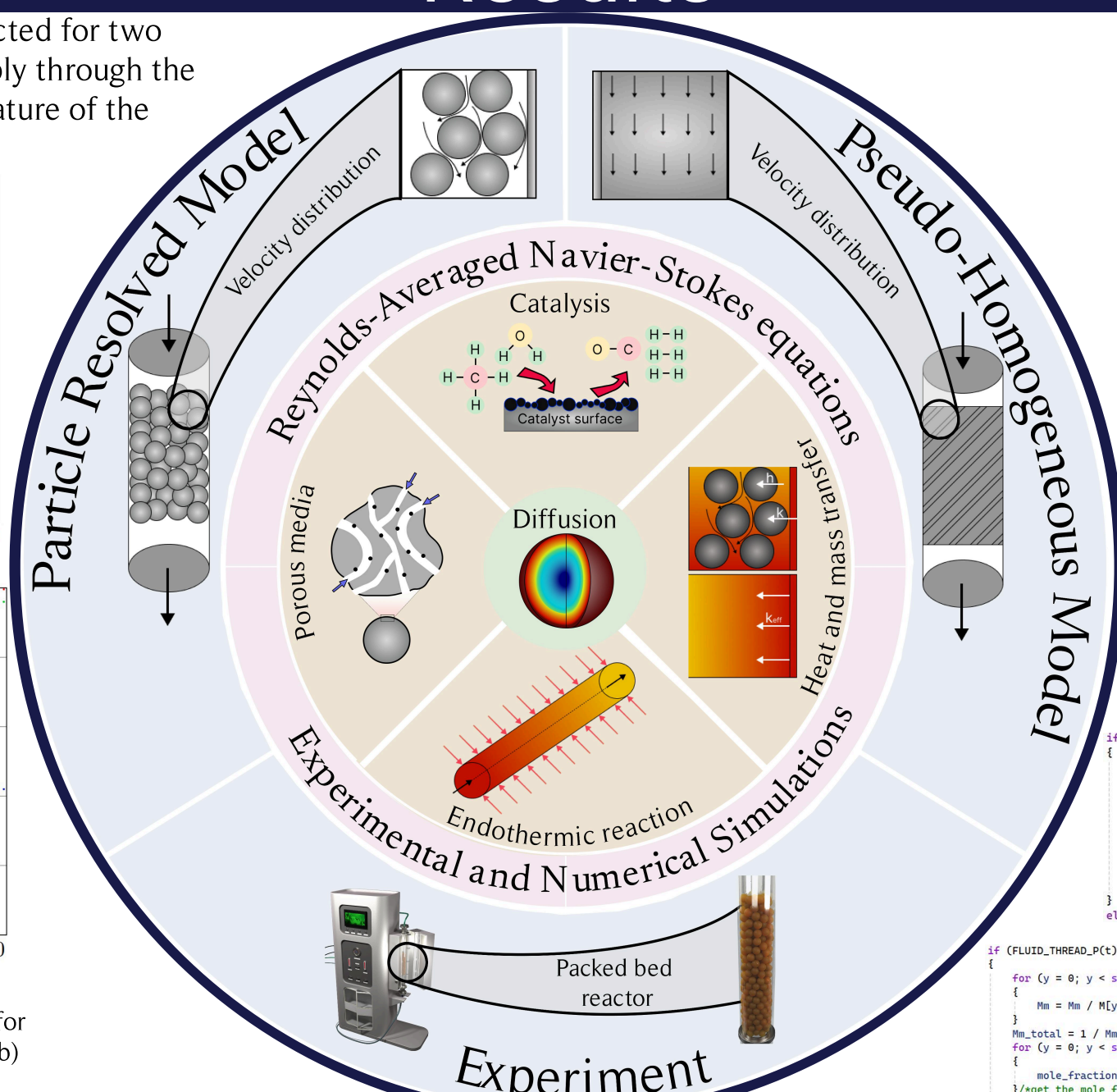
$$\eta = f_d(d) \cdot f_T \left(\frac{T}{1000} \right) \cdot f_b \left(\frac{N_{\text{H}_2\text{O}}}{N_{\text{CH}_4}} \right)$$

Results

Numerical calculations were conducted for two cases: no heat supply and heat supply through the wall to maintain a constant temperature of the reformer wall.



The methane conversion vs. residence time for various temperatures without (a) and with (b) heat supply through the wall.

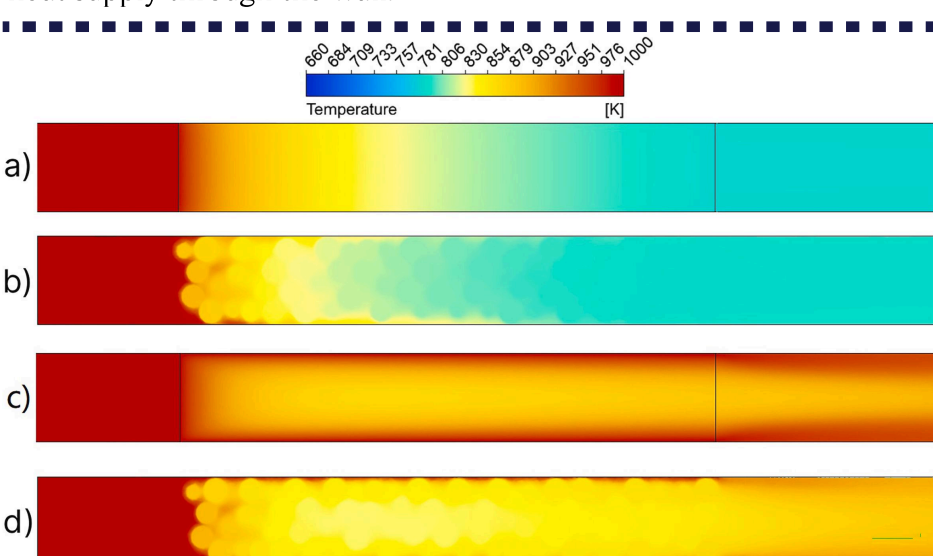


Highlights:

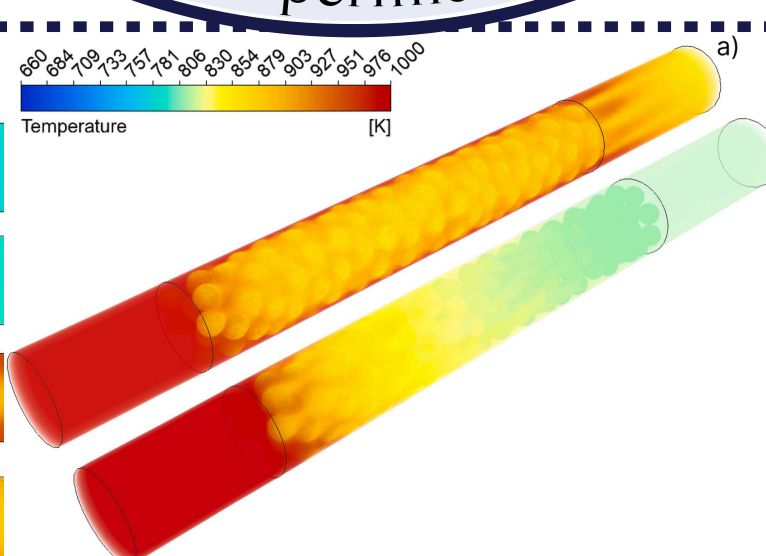
1. Significant deviation in temperature contours for the case with heat supply for PRM and PHM.
2. Good agreement between the results for the case with adiabatic wall.
3. Near-equilibrium methane conversion for the case with heat supply at $T > 1200\text{ K}$.

For chemical kinetics and diffusion model, UDFs were developed in C++ for PHM and PRM.

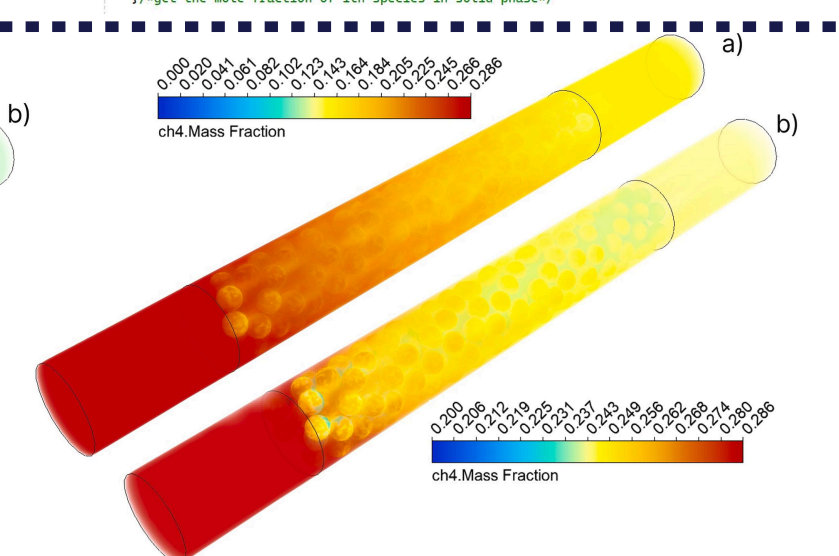
```
if (strcmp(r->name, "reaction-1")
{
    if (FLUID_THREAD_P(t) && THREAD_VAR(t).fluid.porous)
    {
        *rate = 1790. * (k1 * ((P_CH4 * pow(P_H2O, 0.5)) / pow(P_H2, 1.25))
    }
    else
    {
        *rate = 0.;
    }
    *r_t = *rate;
}
else if (strcmp(r->name, "reaction-2")
{
    if (FLUID_THREAD_P(t) && THREAD_VAR(t).fluid.porous)
    {
        for (y = 0; y < species; ++y)
        {
            Mm = Mm / M[y];
        }
        Mm_total = 1 / Mm; /*calculate the average relative molecular mass*/
        for (y = 0; y < species; ++y)
        {
            mole_fraction[y] = (M[y] / Mm_total);
        }
        /*get the mole fraction of ith species in solid phase*/
    }
}
```



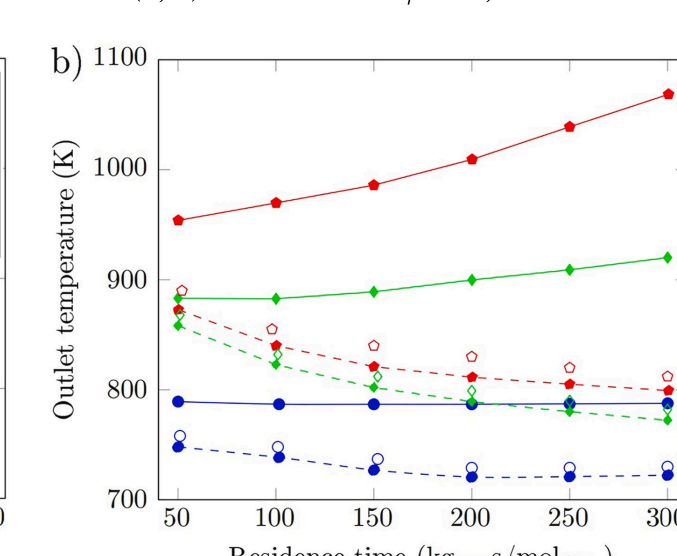
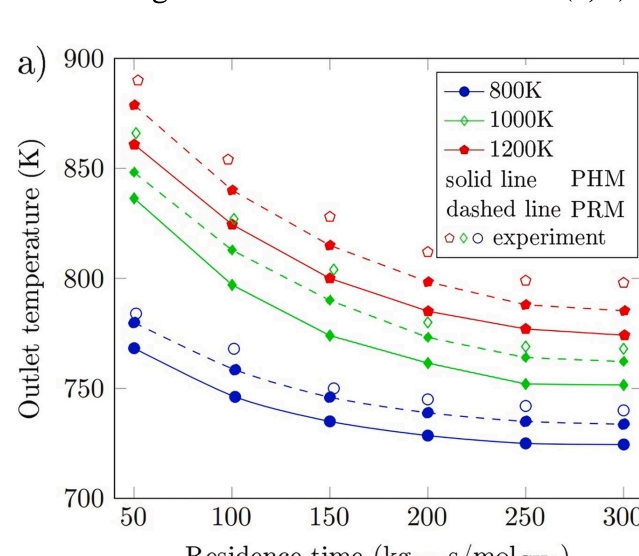
The temperature contours for the cases without (a, b) and with (c, d) heat supply through the wall at $T = 1000\text{ K}$, $\beta = 2$, residence time of $200\text{ kg}_{\text{cat}}/\text{s}/\text{kmol}_{\text{CH}_4}$ obtained via PHM (a, c) and PRM (b, d).



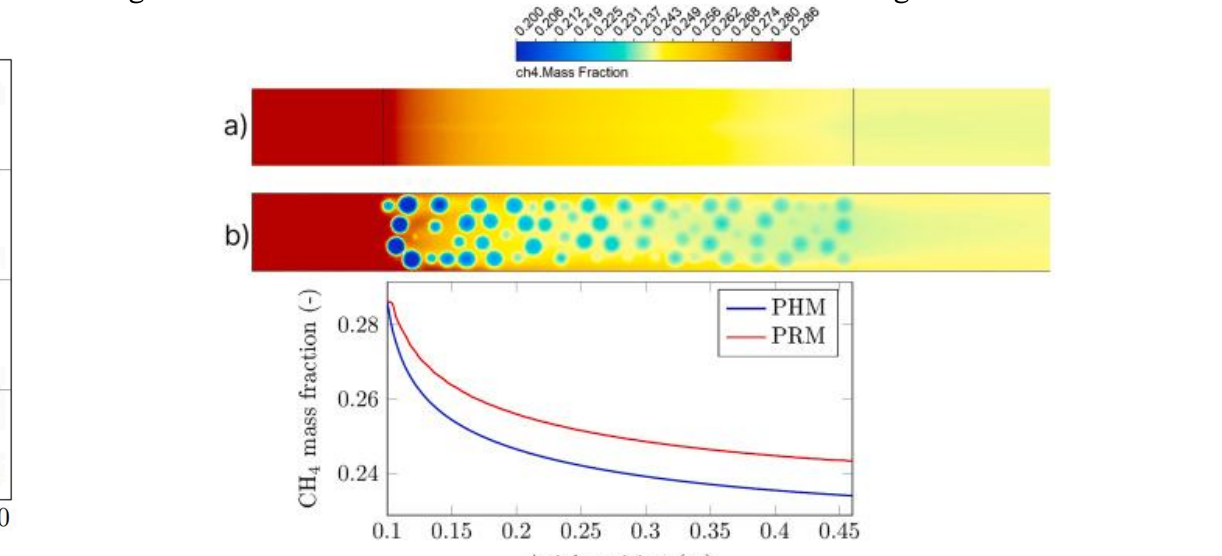
Temperature contours for the reformer with (a) and without (b) heat supply through the wall at $T = 1000\text{ K}$, $\beta = 2$, residence time of $200\text{ kg}_{\text{cat}}/\text{s}/\text{kmol}_{\text{CH}_4}$.



The contours of CH_4 mass fraction for the cases with heat supply (a) and no heat supply (b) at $T = 1000\text{ K}$, $\beta = 2$, and a residence time of $200\text{ kg}_{\text{cat}}/\text{s}/\text{kmol}_{\text{CH}_4}$.



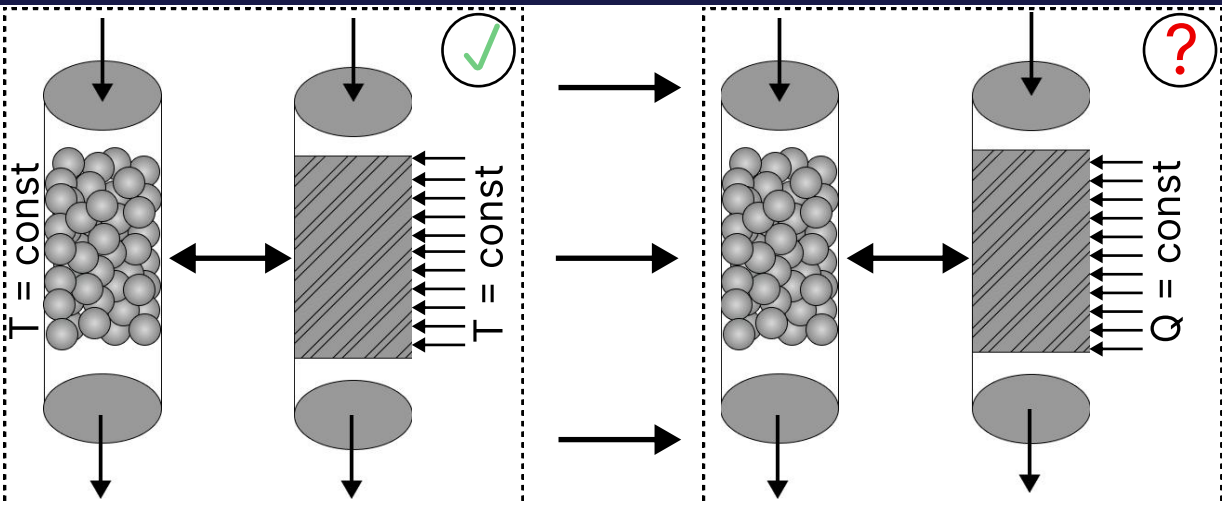
Temperature at the reformer outlet vs. residence time for various inlet and wall temperatures: (a) without heat supply through the wall; (b) with heat supply through the wall. Solid line – pseudo homogeneous model; dashed line – particle resolved model; hollow points – experimental data.



The contours of methane mass fraction for PHM (a) and PRM (b) without heat supply through the wall of reformer filled with spherical particles at $T = 1000\text{ K}$, $\beta = 2$, residence time of $200\text{ kg}_{\text{cat}}/\text{s}/\text{kmol}_{\text{CH}_4}$. Bottom: axial distribution of CH_4 mass fraction for PHM and PRM.

Conclusion and future work

1. PHM requires significantly lower computational resources compared to PRM
2. A strong agreement was observed between the numerical results (CH_4 conversion and outlet temperature) obtained from both models and the experimental data for the case with an adiabatic reformer wall (without heat supply through the wall). For a residence time of $200\text{--}300\text{ kg}_{\text{cat}}/\text{s}/\text{kmol}_{\text{CH}_4}$, the deviation between the results was less than 10%.
3. PHM can be used for numerical simulation of SMR without heat supply through the wall to obtain accurate results.
4. PHM should be refined using an expression for effective thermal conductivity, as the standard parallel model does not accurately predict this parameter. Conversely, PRM effectively captures the performance of the steam methane reforming process.



Further investigation into cases with constant heat flux through the wall is essential for a deeper understanding of the differences between PHM and PRM.